THE COLLECTIONS OF PH COVERINGS DEFINE SITES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [2864, 2868] and proof(s) [[2870, 2882]]; this spec covers the complete environment instead (statement 2864-2868, proof span(s) [(2870, 2882)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Source reference. Original lemma 021C: sources/stacks-chow/tags.txt, source lines 2665–2665. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00VH: sources/stacks-chow/tags.txt, source lines 1147–1147. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0DBI: sources/stacks-chow/tags.txt, source lines 16698–16698. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let Sch_{ph} be a big ph site containing S. Then $(Aff/S)_{ph}$ is a site.*

Proof. Reasoning as in the proof of Lemma 021C it suffices to show that the collection of finite ph coverings $\{U_i \to U\}$ with U, U_i affine satisfies properties (1), (2) and (3) of Sites, Definition 00VH. This is clear since for example, given a finite ph covering $\{T_i \to T\}_{i\in I}$ with T_i, T affine, and for each i a finite ph covering $\{T_{ij} \to T_i\}_{j\in J_i}$ with T_{ij} affine , then $\{T_{ij} \to T\}_{i\in I, j\in J_i}$ is a ph covering (Lemma 0DBI), $\bigcup_{i\in I} J_i$ is finite and each T_{ij} is affine. □

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